NETFLIX EXPLORATORY DATA ANALYSIS (EDA)

Synent Technologies Data Science Internship

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Objective

The objective of this project is to perform Exploratory Data Analysis (EDA) on the Netflix dataset to identify trends, patterns, and meaningful insights through statistical analysis and visualizations.

Dataset Information

Dataset Name: Netflix Dataset

Source: Kaggle

Rows: 8807

Columns: 12

Key Features:

- Type
- Title
- Director
- Country
- Date Added
- Release Year
- Rating
- Duration
- Genre (Listed In)

- Methodology

The following analysis steps were performed:

- Loaded the dataset using Pandas
- Examined dataset structure and missing values
- Generated summary statistics
- Analyzed Movies and TV Shows distribution
- Examined content ratings
- Identified release year trends
- Analyzed top content-producing countries
- Identified popular genres
- Created visualizations to support insights

- Results

The analysis produced the following insights:

- Movies significantly outnumber TV Shows on Netflix.
- Content production increased rapidly after 2010.
- The United States is the leading content-producing country.
- TV-MA is one of the most frequently occurring ratings.
- International Movies, Dramas, and Comedies are among the most popular genres on the platform.

Generated Visualizations:

- Movies vs TV Shows Bar Chart
- Rating Distribution Chart
- Release Trend Graph
- Top Countries Chart
- Top Genres Chart
- Conclusion

Exploratory Data Analysis helped uncover important trends and patterns within the Netflix dataset. The visualizations provided a clear understanding of content distribution, release patterns, ratings, and genre popularity. These insights can support future data-driven decision-making and predictive analytics projects.